Code No: 123AN **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, August/September - 2022
PROBABILITY AND STATISTICS
(Common to ME, CSE, IT, MCT, AME, MIE)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Two dice are thrown and X is a random variable so that $X = Sum(a,b)$ where a and b are numbers appeared on the two dice. Find the probability distribution of X and hence compute its mean and standard deviation.

b) If the probability density of a random variable x is given by

$$f(x) = \begin{cases} (k+2)x^3 , 0 < x < 1 \\ 0 \quad , \text{elsewhere} \end{cases}$$

Find the value k and the probability that the random variable takes on a value (i) greater than 3/4; (ii) between 1/3 and 2/3 . [8+7]

2.a) The probability that a certain kind of component will survive a shock test is 3/4. Find the probability that exactly 2 of the next 4 components tested survive.

b) Find Moment Generating Function for normal distribution and hence find its mean and variance. [5+10]

3.a) A two-dimensional random variable (X,Y) have a joint probability mass function:

$P(X = x, Y = y) = \frac{x^2 + y}{32}$, for $x = 0, 1, 2, 3$ and $y = 0, 1$. Find the marginal distributions of X and Y.

b) The joint probability density function of a two-dimensional random variable (X,Y) is given by: $f(x,y) = \begin{cases} 2 , 0 < x < 1, 0 < y < x \\ 0 , \text{elsewhere} \end{cases}$, find the conditional density functions of Y given $X = x$ and the conditional density functions of X given $Y = y$. [8+7]

4.a) Obtain the rank correlation coefficient for the following data :

x	48	60	72	62	56	40	39	52	30
y	62	78	65	70	38	54	60	32	31

b) Find the regression line of y on x for the following data: [8+7]

x	1	2	3	4	5
y	2	5	3	8	7

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5.a) The means of two large samples of 1000 and 2000 members are 67·5 inches and 68·0 inches respectively. Can the samples be regarded as drawn from the same population having standard deviation 2·5 inches? Test at 5% level of significance.

b) Random samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 200 men and 325 women were in favour of the proposal. Test the hypothesis that proportions of men and women in favour of the proposal, are same against that they are not at 5% level. [8+7]

6.a) Define the Snedecor's F- distribution and write its properties.

b) The Edison Electric Institute has published figures on the number of kilowatt hours used annually by various home appliances. It is claimed that a vacuum cleaner uses an average of 46 kilowatt hours per year. If a random sample of 12 homes included in a planned study indicates that vacuum cleaners use an average of 42 kilowatt hours per year with a standard deviation of 11.9 kilowatt hours, does this suggest at the 0.05 level of significance that vacuum cleaners use, on average, less than 46 kilowatt hours annually? Assume the population of kilowatt hours to be normal. [8+7]

7.a) Explain briefly the main characteristics of Queuing system.

b) A self-service canteen employs one cashier at its counter. 8 customers arrive per every 10 minutes on an average. The cashier can serve on average one per unit. Assuming that the arrivals are Poisson and the service time distribution is exponential. Determine

i) The average number of customers in the system

ii) The average queue length

iii) Average time a customer spends in the system

iv) Average waiting time of each customer. [8+7]

8.a) Explain Stationary and non-stationary random process.

b) Explain Markov process and Markov chain. [7+8]

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